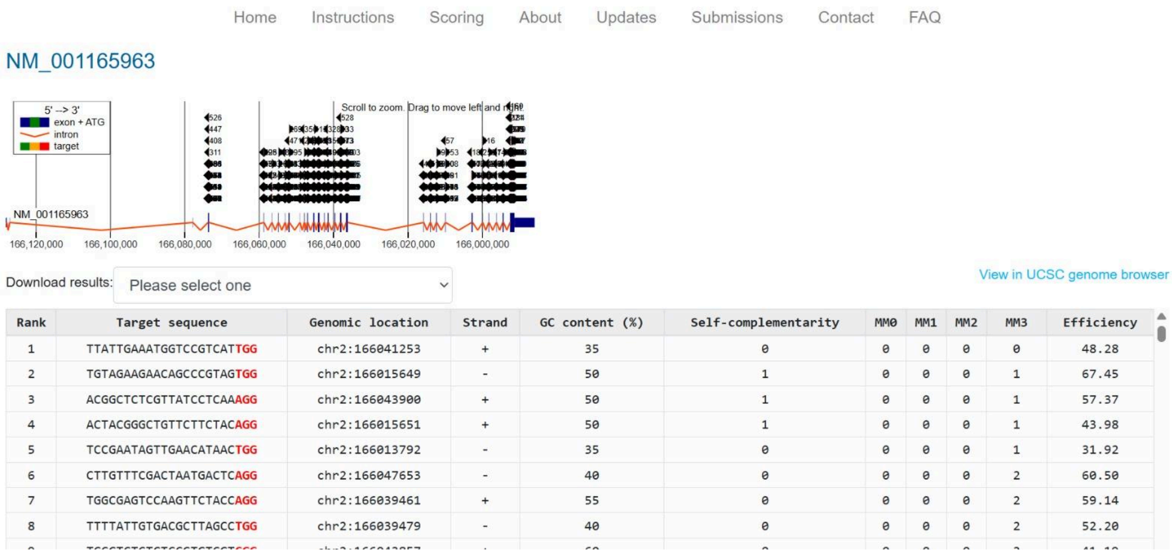
Perform CRISPR guide RNA design to any target gene of interest using any tool

The SCN1A gene was selected as the target because it is associated with epilepsy. Guide RNA design was performed using the CHOPCHOP v3 web server. RefSeq transcript was used to identify the target(NM_001165963 for the gene SCN1A), the organism was Homo sapiens, and the CRISPR system selected was SpCas9 for gene knock-out. Candidate guide RNAs were ranked based on predicted efficiency, GC content, self-complementarity, and off-target potential.



Rank	Target sequence	Genomic location	Strand	GC content (%)	Self-complementarity	MM0	MM1	MM2	MM3	Efficiency
1	TTATTGAAATGGTCCGTCATTGG	chr2:166041253	+	35	0	0	0	0	0	48.28
2	TGTAGAAGAACAGCCCGTAGTGG	chr2:166015649	-	50	1	0	0	0	1	67.45
3	ACGGCTCTCGTTATCCTCAAAGG	chr2:166043900	+	50	1	0	0	0	1	57.37
4	ACTACGGGCTGTTCTTCTACAGG	chr2:166015651	+	50	1	0	0	0	1	43.98
5	TCCGAATAGTTGAACATAACTGG	chr2:166013792	-	35	0	0	0	0	1	31.92
6	CTTGTTTCGACTAATGACTCAGG	chr2:166047653	-	40	0	0	0	0	2	60.50
7	TGGCGAGTCCAAGTTCTACCAGG	chr2:166039461	+	55	0	0	0	0	2	59.14
8	TTTTATTGTGACGCTTAGCCTGG	chr2:166039479	-	40	0	0	0	0	2	52.20

Result

CHOPCHOP generated several candidate guide RNAs targeting the SCN1A gene. Guide RNA rank 2 was selected because it had the highest predicted efficiency score (67.45), an optimal GC content of 50%, low self-complementarity, and minimal predicted off-target activity. These properties indicate that the selected guide is suitable for efficient and specific CRISPR-Cas9-mediated genome editing.